



Indian Institute of Technology Jodhpur
Department of Mathematics
Mathematics II MAL1020
Problem Sheet V

AY 25-26

Semester II

1. Let $x = (1, -2, 3)$ and $y = (4, 0, -1)$ in $\mathbb{R}^3$ with the standard inner product.
 - (a) Compute $\langle x, y \rangle$.
 - (b) Find $|x|$ and $|y|$.
 - (c) Find the angle θ between x and y .

2. Let $u = (2, -1, 1)$ and $v = (1, 2, -3)$ in $\mathbb{R}^3$.
 - (a) Determine whether u and v are orthogonal.
 - (b) Find a scalar $k \in \mathbb{R}$ such that

$$\langle u, v + k(1, 1, 1) \rangle = 0.$$

3. Define an inner product on $\mathbb{R}^3$ by

$$\langle x, y \rangle = 2x_1y_1 + 3x_2y_2 + x_3y_3.$$

Let $x = (1, 1, 1)$ and $y = (2, -1, 3)$.

- (a) Compute $\langle x, y \rangle$.
 - (b) Compute the norm $|x|$ induced by this inner product.
 - (c) Check whether x and y are orthogonal.
4. Let $x = (3, 4)$ and $y = (1, -2)$ in $\mathbb{R}^2$.
 - (a) Compute $|x + y|$ and $|x - y|$.
 - (b) Verify the parallelogram identity:

$$|x + y|^2 + |x - y|^2 = 2(|x|^2 + |y|^2).$$

5. Let $u = (1, 2, 2)$ and $v = (2, 0, 1)$.
 - (a) Compute the orthogonal projection of u onto v :

$$\text{proj}_v(u).$$

- (b) Compute $|\text{proj}_v(u)|$.
 - (c) Verify that the vector $u - \text{proj}_v(u)$ is orthogonal to v .
6. Let $C[0, 1]$ be equipped with the inner product

$$\langle f, g \rangle = \int_0^1 f(x)g(x), dx.$$

Let $f(x) = x$ and $g(x) = x^2$.

- (a) Compute $\langle f, g \rangle$.
- (b) Compute $|f|$ and $|g|$.

7. Let

$$A = \begin{bmatrix} 1 & 2 & 0 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & -1 & 2 & 4 \end{bmatrix}.$$

The Frobenius inner product is defined by

$$\langle A, B \rangle = \sum_{i,j} a_{ij} b_{ij}.$$

- (a) Compute $\langle A, B \rangle$.
 - (b) Compute the Frobenius norm $|A|_F$.
 - (c) Determine whether A and B are orthogonal.
8. Let $x = (1, 2, -1)$ and $y = (2, 0, 3)$.
- (a) Compute $|\langle x, y \rangle|$.
 - (b) Compute $|x|, |y|$.
 - (c) Verify the Cauchy–Schwarz inequality.

9. Let

$$G = \begin{bmatrix} 2 & 1 & 1 & 3 \end{bmatrix},$$

and define

$$\langle x, y \rangle = x^T G y.$$

- (a) Compute $\langle (1, 2), (3, 4) \rangle$.
- (b) Compute the norm $|(1, 2)|$ induced by this inner product.
- (c) Verify that G is positive definite.

10. Let $v = (3, 1)$ and let

$$W = \text{span}(1, 1).$$

- (a) Compute the orthogonal projection of v onto W .
- (b) Compute the distance from v to the subspace W .

11. Let $p_1(x) = a + bx$ and $p_2(x) = c + dx$ be vectors in $\mathcal{P}_1$. Determine a mapping $\langle p_1, p_2 \rangle$ that defines an inner product on $\mathcal{P}_1$.

12. Consider the vector space $\mathbf{R}^2$. Define the mapping $\langle \cdot, \cdot \rangle$ by

$$\langle \mathbf{v}, \mathbf{w} \rangle = 2v_1w_1 + v_1w_2 + v_2w_1 + 2v_2w_2$$

for all vectors $\mathbf{v} = (v_1, v_2)$ and $\mathbf{w} = (w_1, w_2)$ in $\mathbf{R}^2$